

Problem

Find the least positive integer m such that

$$\binom{2n}{n} < m^n$$

for all positive integers n .

Source: Romania 1990

§1 Harnessing the Binomial Expansion

The central binomial coefficient is one of the terms appearing in the expansion

$$(1 + 1)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k}.$$

Since all terms of this sum are positive, we have

$$\binom{2n}{n} < \sum_{k=0}^{2n} \binom{2n}{k} = (1 + 1)^{2n} = 4^n.$$

Thus the inequality

$$\binom{2n}{n} < 4^n$$

holds for every positive integer n . Hence $m = 4$ is a valid choice.

§2 Showing Minimality

It remains to prove that no smaller positive integer works.

The only candidate smaller than 4 is $m = 3$. Taking $n = 5$, we obtain

$$\binom{10}{5} = 252,$$

while

$$3^5 = 243.$$

Therefore

$$\binom{10}{5} > 3^5,$$

so the inequality fails when $m = 3$.

Consequently, no integer $m < 4$ satisfies the required condition for all positive integers n .

§3 The Final Conclusion

The least positive integer satisfying

$$\binom{2n}{n} < m^n$$

for every positive integer n is

$$\boxed{4}.$$